Experimental Report:

Graph-Based Retrieval-Augmented Generation

1. System Configuration

This project compares a **Graph RAG** pipeline with a **Vector RAG** baseline using *Attention Is All You Need* as the sole knowledge source.

The configured models were:

- **Knowledge Graph construction:** gpt-4o through KGGen
- **Answer generation:** gpt-4o
- **Embedding model:** text-embedding-3-small

Both systems use the default generation parameters:

- **Temperature** = 0.1
- **Top-p** = 0.3

The original experimental data are stored in:

results/ Experimental_Validation_A.txt

results/ Experimental_Validation_B.txt

2. Experimental Validation A: Graph RAG vs Vector RAG

Five questions were tested on both systems.

Query	Graph RAG	Vector RAG
Q1	Partially correct. Retrieved the affiliation triples for three authors, but the final answer mentioned only two.	Correctly identified Ashish Vaswani, Noam Shazeer and Łukasz Kaiser.

Query	Graph RAG	Vector RAG
Q2	It retrieved that residual dropout applies to sub-layers, but the numerical rate value was absent from the KG.	Correctly returned 0.1.
Q3	Retrieved the dataset but did not link it to the translation task. The number of sentence pairs was also absent from the KG.	Correctly returned the WMT 2014 English-German dataset and approximately 4.5 million sentence pairs.
Q4	Could not identify the techniques or parameter values from the retrieved triples.	Also failed because the retrieved chunks did not contain sufficient information.
Q5	Retrieved relevant triples and produced a partially correct answer including parallelization and long-range dependencies. Computational complexity was retrieved but not considered as an advantage, while interpretability was included into the answer.	Produced a plausible synthesis based on parallelization, modeling distant dependencies, and performance. Computational complexity was also omitted.

Comparison and Justification

Vector RAG was generally more reliable for fact-oriented questions because retrieving original text chunks preserved quantitative and contextual details that were sometimes lost during KG construction. In contrast, Graph RAG showed some limitations. First, knowledge graph construction loss occurred when KGGen failed to represent information from the source paper, especially numerical values or some relationship information. In such cases, no retrieval strategy could recover the missing facts. Second, graph retrieval limitations appeared when relevant information existed in the KG but was weakly connected or disconnected from the selected seed entities, preventing traversal from reaching it. Vector RAG avoids much of the construction-loss problem by retrieving directly from the source text, although it can still fail when relevant chunks are not selected. There were also issues of imperfect generation from retrieved triples, for example, for Q1, the third author was retrieved but omitted from the final answer.

3. Experimental Validation B: Parameter Analysis

With one specific query, the retrieved evidence was kept effectively constant while changing only the generation profiles.

Graph RAG

Correctly identified: multi-headed self-attention; position-wise fully connected feed-forward network; residual connection; layer normalization.

Profiles 1 and 2 were almost identical and additionally mentioned residual dropout. Profile 3 omitted residual dropout and gave a slightly shorter answer focused only on the two sub-layers and the operations around them.

Vector RAG

Correctly identified the same information, and the explicit formula LayerNorm(x + Sublayer(x)) from the retrieved text.

Profiles 1 and 2 produced identical answers. Profile 3 contained the same factual content but used slightly different wording and presentation, including explicitly referring to the Transformer model and the formula.

Parameter Analysis

Across all three parameter profiles, the responses remained highly stable. Although temperature and top-p control sampling diversity, their observable effect was limited in this experiment. One likely reason is the strongly constrained RAG prompt: the models were required to use only the retrieved evidence, avoid unsupported claims, and provide concise factual answers. In addition, the query has a narrowly defined factual answer, and the retrieved evidence remained effectively constant across profiles. These constraints substantially reduced the space of valid responses. Higher temperature and top-p therefore produced mainly minor stylistic differences rather than changes in factual content. This suggests that sampling parameters may have a relatively limited effect when generation is tightly grounded by fixed retrieval context and strict factuality instructions.